# Calculation questions – 4.7 Magnetism and electromagnetism

AQA GCSE Physics (8463) Higher · Paper 2 · 10 questions · 38 marks · answers at the back

## Equations for this topic

| Equation | Symbols and units | Exam |
|---|---|---|
| F = BIl (HT) | F (N), B (T), I (A), l (m) | On sheet |
| Vp/Vs = np/ns (triple, HT) | V (V), n (turns) | On sheet |
| Vp × Ip = Vs × Is (triple) | V (V), I (A); 100% efficient | On sheet |
| P = VI and P = I²R | P (W), V (V), I (A), R (Ω) | Learn it |
| W = mg | W (N), m (kg), g (N/kg) | Learn it |

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

## Grade 4–5

**1** A 0.50 m wire carrying 3.0 A is at right angles to a 0.20 T magnetic field. Calculate the force on the wire. [2 marks]

Answer: _____ Unit: _____

**2** (Triple) A transformer has 1000 turns on the primary coil and 50 on the secondary. The primary p.d. is 230 V. Calculate the secondary p.d. [3 marks]

Answer: _____ Unit: _____

## Grade 6–7

**3** An 8.0 cm wire carrying 2.5 A experiences a force of 0.030 N in a magnetic field. Calculate the magnetic flux density. [3 marks]

Answer: _____ Unit: _____

**4** (Triple) A step-up transformer changes 25 kV to 400 kV. The primary coil has 2000 turns. Calculate the number of turns on the secondary coil. [3 marks]

Answer: _____ Unit: _____

**5** (Triple) A 100% efficient transformer has a primary p.d. of 230 V and primary current 0.50 A. The secondary p.d. is 12 V. Calculate the secondary current. [3 marks]

Answer: _____ Unit: _____

**6** A 0.20 m wire is placed in a 0.050 T field. Calculate the current needed to produce a force of 0.040 N. [3 marks]

Answer: _____ Unit: _____

## Grade 8–9

**7** (Triple) A phone charger transformer steps 230 V down to 5.0 V. The secondary coil has 100 turns and supplies 2.0 A. (a) Calculate the number of primary turns. (b) Calculate the primary current if it is 100% efficient. (c) Calculate the primary current if it is 90% efficient. [6 marks]

Answer: _____ Unit: _____

**8** A motor coil has 50 turns. Each side of the coil has 0.040 m of wire in a 0.30 T field and carries 1.5 A. (a) Calculate the total force on one side of the coil. (b) The sides are 0.030 m from the axle. Calculate the total turning moment on the coil. [5 marks]

Answer: _____ Unit: _____

**9** A horizontal wire 0.25 m long with a mass of 5.0 g sits at right angles to a horizontal 0.40 T field. Calculate the current needed for the magnetic force to support the wire's weight. [4 marks]

Answer: _____ Unit: _____

**10** (Triple) A power station produces 200 MW at 25 kV. A transformer steps this up to 400 kV. (a) State the turns ratio. (b) Calculate the current in the transmission cables. (c) The cables have a resistance of 2.0 Ω. Calculate the power lost as a percentage of the power transmitted. [6 marks]

Answer: _____ Unit: _____

# Answers – 4.7 Magnetism and electromagnetism

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

| Q | Working | Answer |
|---|---|---|
| 1 | F = 0.20 × 3.0 × 0.50 | **0.30 N** |
| 2 | Vs = 230 × 50 ÷ 1000 | **11.5 V** |
| 3 | B = 0.030 ÷ (2.5 × 0.080) | **0.15 T** |
| 4 | ns = 2000 × 400 ÷ 25 | **32 000 turns** |
| 5 | 230 × 0.50 = 12 × Is | **9.6 A** |
| 6 | I = 0.040 ÷ (0.050 × 0.20) | **4.0 A** |
| 7 | (a) np = 100 × 230 ÷ 5.0 (b) Pout = 5.0 × 2.0 = 10 W; Ip = 10 ÷ 230 (c) Pin = 10 ÷ 0.90 = 11.1 W; Ip = 11.1 ÷ 230 | **(a) 4600 (b) 0.043 A (c) 0.048 A** |
| 8 | (a) per turn 0.30 × 1.5 × 0.040 = 0.018 N; × 50 (b) two sides: 2 × 0.90 × 0.030 | **(a) 0.90 N (b) 0.054 Nm** |
| 9 | W = 0.0050 × 9.8 = 0.049 N; I = 0.049 ÷ (0.40 × 0.25) | **0.49 A** |
| 10 | (a) 25 : 400 (b) I = 200×10⁶ ÷ 400×10³ (c) P = 500² × 2.0 = 500 000 W; 0.50 MW ÷ 200 MW | **(a) 1 : 16 (b) 500 A (c) 0.25%** |

**Common mistakes.** Not converting cm and g (Q3, Q9); flipping the turns ratio (step-down means fewer secondary turns); forgetting the force acts on every turn of a coil (Q8).